

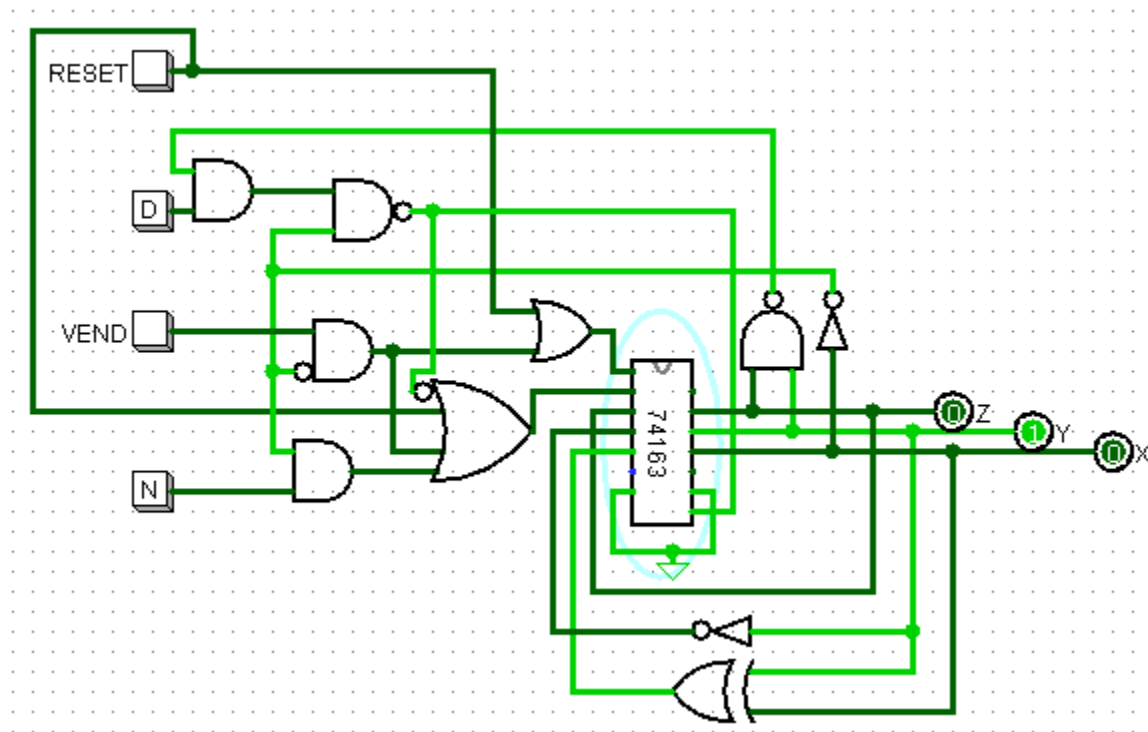
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Studio C

Vending Machine

- Only Nickels and Dimes
- Maximum is 20 cents
- 5 States - 0 cents, 5 cents, 10 cents, 15 cents, 20 cents
- If not a nickel and not a dime is entered, then no transition occurs,
- Moore Notation

Logisim Simulation



Truth Table:

000 - 0.00

001 - 0.05

010 - 0.10

011 - 0.15

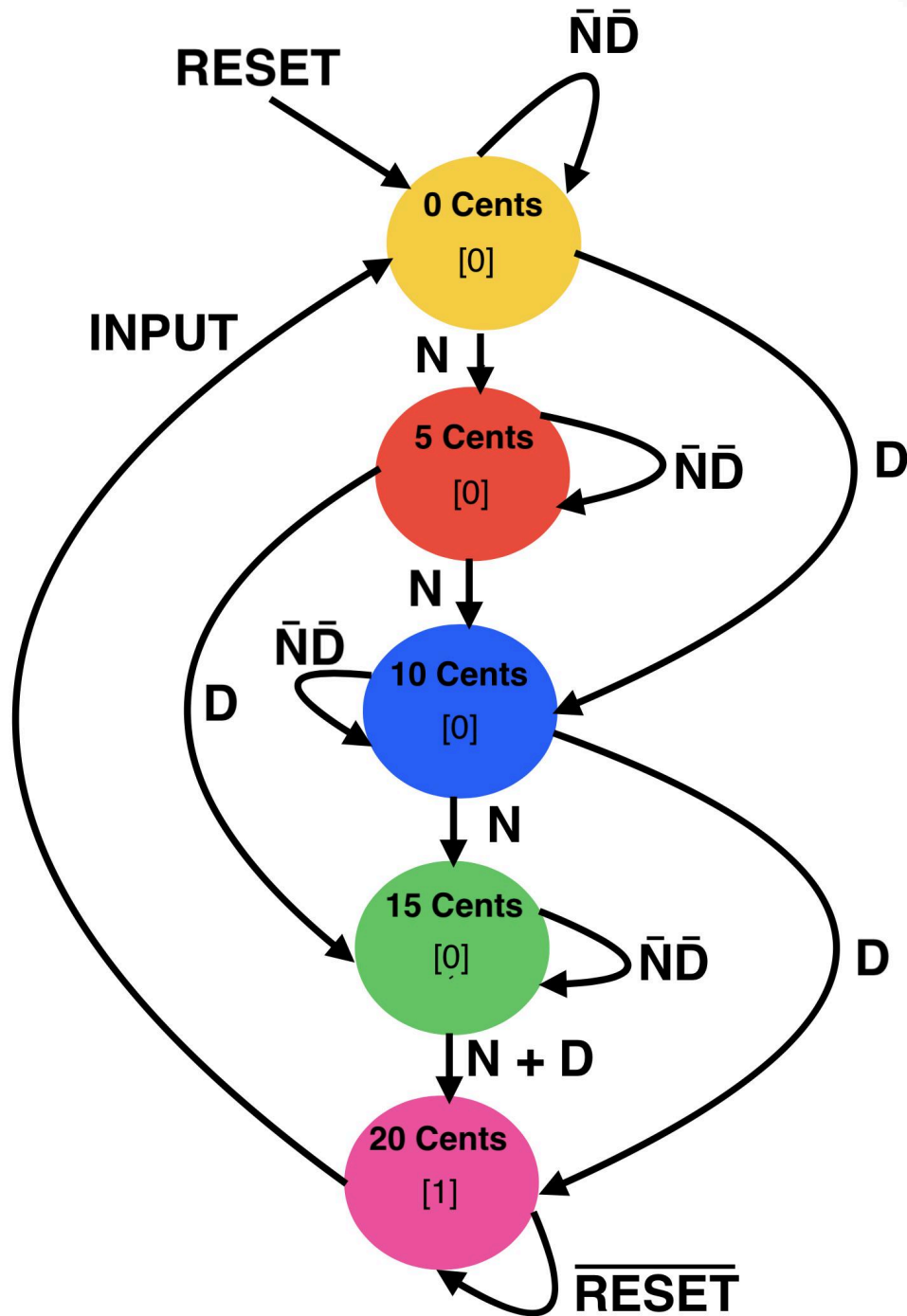
100 - 0.20

For Reset = 0

D	N	Input (Vend item)	X	Y	Z	X ⁺	Y ⁺	Z ⁺	Output
0	0	X	0	0	0	0	0	0	0
0	1	X	0	0	0	0	0	1	0
1	0	X	0	0	0	0	1	0	0
0	0	X	0	0	1	0	0	1	0
0	1	X	0	0	1	0	1	0	0
1	0	X	0	0	1	0	1	1	0
0	0	X	0	1	0	0	1	0	0
0	1	X	0	1	0	0	1	1	0
1	0	X	0	1	0	1	0	0	0
0	0	X	0	1	1	0	1	1	0
0	1	X	0	1	1	1	0	0	0
1	0	X	0	1	1	0	1	0	0
0	0	0	1	0	0	1	0	0	1
0	0	1	1	0	0	0	0	0	0
0	1	0	1	0	0	1	0	0	1
0	1	1	1	0	0	0	0	0	0
1	0	0	1	0	0	1	0	0	1
1	0	1	1	0	0	0	0	0	0

For Reset = 1, always lead to 000 state.

Complete State Diagram



Real World Function: Tracking the amount of change entered into a vending machine, or similar styled machine, and dispensing the product when enough change has been entered.

Physical Circuit

